FLUOROURACIL

DPD deficiency testing
Mucositis & diarrhoea

CHEMICAL NAME 5-fluoropyrimidine-2,4(1H,3H)-dione (5-fluorouracil)
CLASS Antimetabolite (pyrimidine analogue) cytotoxic
TRADE NAMES Adrucil · Fluorouracil injection · Efudex (5% cream) · Carac

THERAPEUTIC USES / INDICATIONS

- Colorectal cancer (FOLFOX, FOLFIRI), gastric, oesophageal, pancreatic, breast, head and neck cancers.
- Topical: actinic keratosis, superficial basal cell carcinoma.

PHARMACODYNAMICS

- Converted to **FdUMP** → inhibits **thymidylate synthase** → ↓ DNA synthesis; also incorporates into RNA.
- Folinic acid stabilises FdUMP–enzyme complex (↑ effect).

ADMINISTRATION

- **IV** bolus and/or continuous infusion (46 h via ambulatory pump/PICC) — protocol-based.
- Often with folinic acid (leucovorin) to enhance effect.
- Topical cream twice daily for 2–4 weeks (expect inflammation).
- DPD (DPYD) genotype/phenotype testing before first dose

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Mucositis, stomatitis, diarrhoea.
- Nausea, fatigue.
- Hand–foot syndrome (palmar-plantar erythrodysaesthesia) with infusions; photosensitivity.

ADVERSE EFFECTS (SERIOUS)

- Severe/fatal toxicity in DPD deficiency (mucositis, diarrhoea, neutropenia, neurotoxicity).
- Myelosuppression (bolus).
- Coronary vasospasm, angina, MI.
- Hyperammonaemic encephalopathy; cerebellar ataxia.

DOSAGES

- **FOLFOX/FOLFIRI (modified de Gramont):** 400 mg/m² IV bolus, then 2,400 mg/m² infusion over 46 h every 2 weeks.
- **With radiotherapy:** 200–225 mg/m²/day continuous infusion.
- **DPD partial deficiency:** reduce dose by ~50%; complete deficiency: avoid.

CONTRAINDICATIONS & CAUTIONS

- Complete DPD deficiency.
- Use with brivudine/sorivudine (within 4 weeks) — fatal DPD inhibition.
- Severe myelosuppression; pregnancy, breastfeeding.

KEY INTERACTIONS

- Brivudine: contraindicated (fatal).
- **Warfarin:** ↑ INR (monitor).
- **Phenytoin:** ↑ levels.
- **Folinic acid:** ↑ efficacy and toxicity.

PHARMACOKINETICS

- Rapidly metabolised (80%) by dihydropyrimidine dehydrogenase (DPD) in liver.
- **t½:** 8–20 min — hence continuous infusion.

NURSING MANAGEMENT

- Confirm DPD testing result before first cycle.
- Pump care: check line, pump function, disconnect at 46 h; spill kit at home.
- Oral care regimen; assess mouth, stools (≥4 above baseline → report), hands/feet.
- Report chest pain immediately (stop infusion).
- FBC before each cycle.

PATIENT EDUCATION

- Report diarrhoea (4+ extra stools/day), mouth sores, chest pain or fever urgently.
- Keep the pump close; do not get it wet; know what to do with spills.
- Moisturise hands/feet; avoid sun.
- Topical cream: expect redness and crusting; avoid eyes.

KEY POINTS

- Pyrimidine antimetabolite; thymidylate synthase inhibitor.
- Mucositis, diarrhoea, hand–foot syndrome.
- DPD deficiency → fatal toxicity; test first.
- Coronary vasospasm; never with brivudine.